



India Metro Air Quality Forecasting

Next-day PM2.5 modelling for Delhi, Mumbai, and Hyderabad

Project summary	Verified result
Data	3,747 city-day records from August 2022 to December 2025
Model	Ridge regression with alpha 10 and chronological validation
Final test	MAE 8.50 ug/m3, R2 0.851, alert recall 0.871
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Executive brief

This project estimates the next day's city level mean PM2.5 using current and historical pollution, weather, city, and seasonal signals. It is a compact machine learning study rather than a production alerting system.

Decision	Evidence
Selected model	Ridge alpha 10, chosen on late-2024 validation MAE
Final test	1,095 unseen city-day rows from calendar year 2025
Accuracy	MAE 8.50 ug/m3 and R2 0.851
Baseline	Persistence MAE 8.92 ug/m3
Alert view	Recall 0.871 at 60 ug/m3

Project value

- Prevents future information from leaking into training.
- Compares a learned model with a strong persistence baseline.
- Reports city-level error instead of hiding it behind one aggregate score.
- Connects model evidence to a responsible-use boundary.

Problem framing and stakeholders

The analytical question is deliberately narrow: how accurately can a compact model estimate tomorrow's mean PM2.5 for three Indian metros when today's pollution and weather are known? A continuous target preserves more information than a category label.

Role	Question answered
Data scientist	Was leakage prevented and was the baseline credible?
Environmental analyst	Which cities and conditions drive the error?
Project reader	Can the model choice and limitations be explained clearly?
Engineer	Can the data, features, model, and evidence be reproduced?

Out of scope

- Neighbourhood exposure estimates or CPCB regulatory reporting.
- Health advice, emergency alerts, or pollution-causality claims.
- A claim that atmospheric model fields equal audited station observations.

Data, provenance, and analytical grain

Open-Meteo air-quality and historical-weather APIs provide the reproducible source. Pollutants originate from Copernicus Atmosphere Monitoring Service model output. Raw JSON stays local; checksums and a versioned city-day panel provide traceability.

Attribute	Value
Cities	Delhi, Mumbai, Hyderabad
Coverage	2022-08-01 through 2025-12-31
Processed grain	One city and date
Rows	3,747 before feature-window exclusions
Pollutants	PM2.5, PM10, nitrogen dioxide, ozone
Weather	Temperature, humidity, rain, wind speed, wind gust
Timezone	Asia/Kolkata

One representative coordinate per city makes the project reproducible but cannot represent roadside, industrial, residential, and suburban variation. The report carries this limitation into every interpretation.

Feature engineering and leakage control

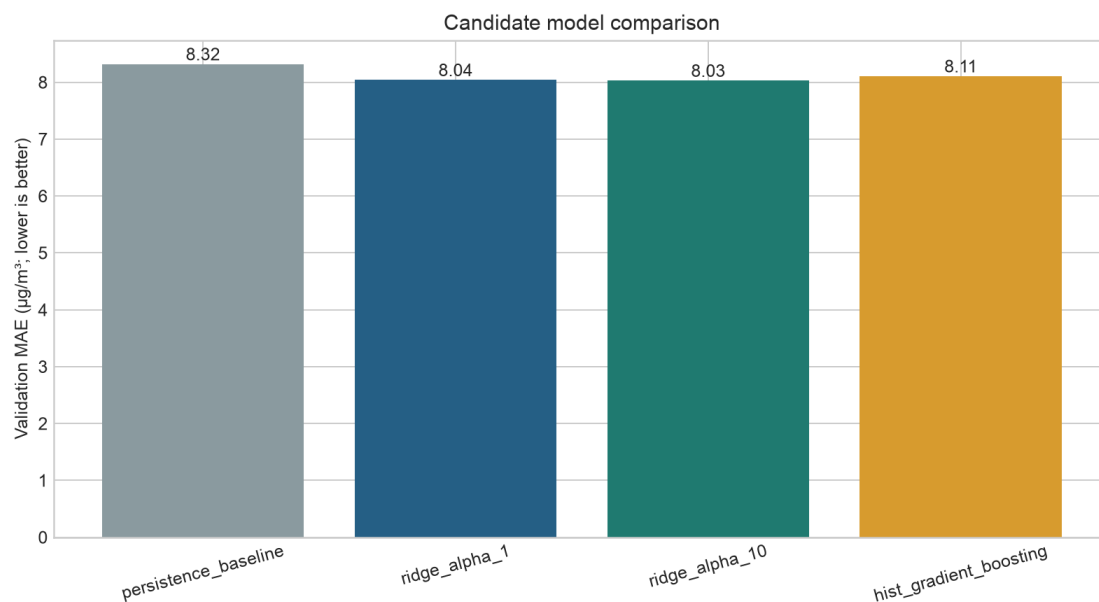
Feature family	Examples	Control
Pollution history	Current, lags, 3-day and 7-day rolls	Grouped within city
Related pollutants	PM10, NO2, ozone	Current day or earlier
Weather	Temperature, humidity, rain, wind	Joined by city and date
Seasonality	Month and day-of-year cycles	Cyclical encoding
Geography	City indicator	One-hot encoding

Pipeline sequence

- Aggregate hourly pollution to a daily city panel.
- Join weather at the same city-date grain.
- Create lags and rolling windows independently within each city.
- Shift the target one day forward within city.
- Split chronologically before model selection or evaluation.

Validation and model selection

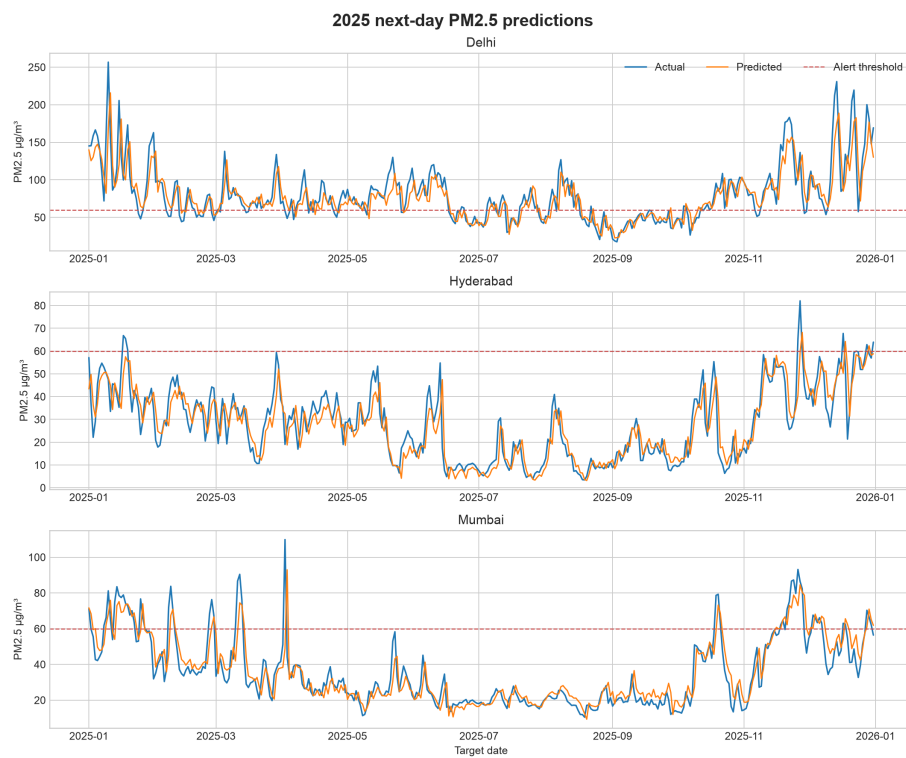
Training ends on 30 June 2024, validation covers July through December 2024, and the final test is calendar year 2025. Candidate choice uses validation MAE only. The selected model is refit on training plus validation data.



Ridge alpha 10 wins by a narrow validation margin. The simpler model improves interpretability without claiming a dramatic advantage over persistence.

Final test performance

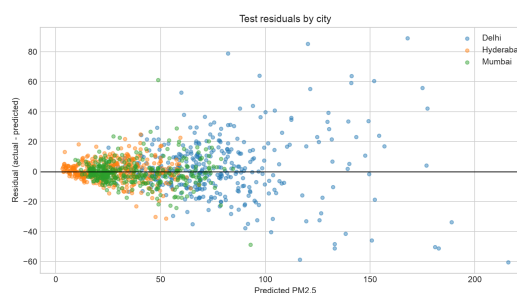
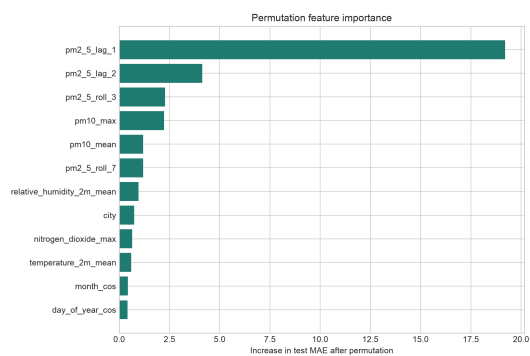
Metric	Ridge	Persistence
MAE	8.50	8.92
RMSE	13.45	14.93
R2	0.851	0.817
Alert recall	0.871	0.845
Alert F1	0.849	0.845



The learned model improves MAE by about 4.7 percent. Persistence remains strong, which is credible for a next-day target dominated by recent conditions.

Explainability and city-level error

City	MAE	R2	Alert F1
Delhi	14.47	0.707	0.883
Mumbai	5.63	0.812	0.774
Hyderabad	5.39	0.795	0.000



Delhi drives most of the remaining absolute error. Hyderabad rarely crosses the threshold in 2025, so its zero alert F1 is exposed rather than hidden behind aggregate performance.

Limitations and responsible interpretation

Limitation	Why it matters	Improvement
Modelled source	Not audited station observations	Add CPCB station data
One point per city	No within-city variation	Build a spatial panel
Realised weather	Cleaner than forecast inputs	Use issued forecasts
No intervals	Point estimates hide risk	Calibrate uncertainty
Sparse alerts	City F1 can be unstable	Use longer backtests

Responsible-use boundary

- Do not use this model for health decisions or emergency warnings.
- Do not present 60 ug/m3 as the complete Indian AQI.
- Do not generalise one-coordinate results to every neighbourhood.
- Treat the output as a reproducible modelling study and baseline comparison.

Reproducibility and evidence

Step	Command or artifact
Environment	<code>uv sync</code>
Acquire	<code>uv run python scripts/download_data.py</code>
Train	<code>uv run python scripts/train_model.py</code>
Verify	<code>uv run pytest -q</code> and <code>uv run ruff check .</code>
Evidence	<code>reports/model_metrics.json</code> and <code>reports/figures/</code>
Documentation	<code>docs/dataset.md</code> , <code>model-card.md</code> , and <code>project-report.md</code>

Skills demonstrated

- API acquisition, checksum provenance, pandas features, and time-aware splits.
- Scikit-learn pipelines, model selection, baseline comparison, and importance.
- Regression and threshold metrics, error analysis, tests, and documentation.

Primary sources

- Open-Meteo Air Quality and Historical Weather APIs.
- Copernicus Atmosphere Monitoring Service attribution through Open-Meteo.
- Central Pollution Control Board national AQI documentation.